Wiley Interdisciplinary Reviews: Nanomedicine and Nanobiotechnology

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ADVANCED REVIEW

Development of Self-Adjuvants in mRNA Vaccine and Its Application in Disease Prevention and Treatment

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Received: 16 May 2024 | **Revised:** 13 September 2024 | **Accepted:** 13 October 2024

Associate Editor: Andrew Wang | **Co-Editor-in-Chief:** Fabiana Quaglia

Funding: This work was supported by Postdoctoral Research Foundation of China (Grant 2024M754162) and Henan Key Science and Technology Research (Grant 242102311235).

Keywords: mRNA vaccine | nanocarriers | self-adjuvants

ABSTRACT

Adjuvants augment the immunogenicity of vaccines when co-administered with messenger RNA (mRNA) antigens. In recent years, nanotechnology and nanoscience have seen significant growth, resulting in the discovery of synthetic small molecule compounds, natural extracts, and nanomaterials with self-adjuvant properties for nano delivery. The materials exhibit robust immune activity and efficiently activate various innate immune signaling pathways. Moreover, they possess a comparatively simple chemical composition in contrast to conventional adjuvants. This significantly streamlines the manufacturing process of vaccine formulations. Therefore, these self-adjuvant materials theoretically improve the reproducibility of adjuvant production and quality control. Herein, this review summarizes the current research and development progress of mRNA adjuvants, with a specific focus on various types of mRNA adjuvants, notably self-adjuvant nanomaterials. It discusses the current research status on a range of diseases and investigates the potential development of mRNA vaccine adjuvants.

1 | Introduction

Vaccination has been widely employed as the most efficacious measure to mitigate the transmission of infectious diseases. For example, the smallpox virus has been successfully eradicated, and the global prevalence of polio, measles, and other infectious diseases has significantly decreased due to the widespread administration of vaccines (Kayser and Ramzan 2021; Zhang et al. 2023). Among the various types of vaccines available, messenger RNA (mRNA) vaccines have gained attention for their high efficacy, safe administration, and cost-efficient production in comparison to conventional vaccines (Chen et al. 2023; Pardi et al. 2018; Xie, Yao, and Xia 2023). In particular, after the successful rapid implementation of mRNA vaccines combined with lipid nanoparticles (LNPs) against SARS-CoV-2, technical challenges in vaccine preparation were overcome. This enabled the rapid and cost-effective production of mRNA vaccines that exhibit excellent stability and high efficacy.

Generally, mRNA vaccines typically consist of a synthetic RNA molecule and an adjuvant component (Kwon et al. 2022; Pulendran, Arunachalam, and O'Hagan 2021; Xie, Yao, and Xia 2023). Indeed, the initial mRNA was not utilized as a treatment due to limitations such as poor stability and efficacy, as well

The first two authors contributed equally to this article.

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as excessive immune stimulation. Fortunately, some delivery vectors, such as LNPs, have been cleverly used to overcome the limitations of mRNA medicine. This has led to advancements in the exploration of mRNA therapy and its applications (Chen et al. 2023; Karandikar et al. 2017). In addition, adjuvants, as the main components of mRNA vaccines, can enhance the strength, speed, and persistence of the immune response to the vaccine (Zhao et al. 2023). Currently, vaccines and adjuvants cannot work simultaneously as two separate components (Facciolà and Visalli 2022). To this end, delivery vectors with self-adjuvant properties undoubtedly provide a robust platform for spatiotemporal and controllable regulation of the interaction between vaccines and immune cells. These delivery systems can easily be equipped with various functions, such as spatiotemporal responsive delivery, targeting lymph nodes, and activating innate immune cells (Rajashri et al. 2022; Roth et al. 2022; Zhu 2021). Some lipid-based delivery vectors have been identified with self-adjuvant properties that effectively trigger innate immune responses and can simplify the vaccine manufacturing process (Liao et al. 2022; Zhang et al. 2021; Zhang, Yang, et al. 2023). In addition, nanomaterials that combine adjuvants with delivery carriers, such as polymers, inorganic materials, biomaterials, and their derivatives, have been widely employed in the development of vaccine carriers in recently years. These carriers are capable of activating innate immune signaling pathways such as STING, TLRs, and NLRs (Luo et al. 2017; Xie et al. 2022). Another noteworthy feature is the relatively simple chemical structure of these self-adjuvant materials, which conceptually enhances the repeatability of adjuvant manufacturing and quality control. In addition, self-adjuvant nanovaccines are composed of nanomaterials with intrinsic immune-stimulating activity. This composition ensures a high degree of spatio-temporal synergism and optimal pharmacokinetic properties of the vaccine and adjuvant without the need for additional adjuvants. These nanovaccines feature immune-enhancing effects that aim to improve the therapeutic effect on tumors (Liu et al. 2021; Luo et al. 2020). For example, Zhang, Yang, et al. (2023) reported that a dendritic polypeptide (KK2DP7) can enhance antigen uptake and cross-presentation efficiency of antigen-presenting cells (APCs) as a delivery carrier and immune adjuvant. Researchers discovered that a cancer vaccine based on KK2DP7, when combined with immune checkpoint blocking therapy, effectively prevented tumor recurrence in postoperative recurrent tumor models.

With the rapid development of mRNA vaccine technology and nanoscience, an increasing number of self-adjuvant nano-delivery carriers have been developed, demonstrating excellent capability in enhancing vaccine immune response. In this review, we have summarized the research status and existing types of mRNA vaccine adjuvants, outlined the research progress of mRNA self-adjuvant vaccines for cancer and other diseases in recent years, and provided an outlook on the potential application of self-adjuvant nanocarriers.

2 | Development of mRNA Vaccines

Although conventional vaccines can offer long-lasting protection against various diseases, they still have limitations that restrict their use in disease prevention and treatment. The production of dendritic cell (DC) vaccines necessitates the isolation and manipulation of the patient's autologous cells, a procedure that is both laborious and time-consuming (Sabado, Balan, and Bhardwaj 2017; Zhang, Tang, et al. 2023). The design and manufacture of microbial-based vaccines require an intricate process (Bambini and Rappuoli 2009; Frost et al. 2023). Peptide vaccines may pose risks of immune evasion due to MHC restriction (Ryan, Jonathan, and Olivia 2019; Zhang, Tang, et al. 2023). DNA vaccines pose potential risks such as genomic alterations, sustained expression over an extended period, and the generation of autoantibodies targeting DNA (Lu et al. 2023). Therefore, it is crucial to select an appropriate vaccine regimen for various types of diseases.

mRNA vaccines are composed of a singular nucleotide sequence serving as a template for protein translation. For instance, the initial documentation of the effective application of in vitro transcription (IVT) mRNA was reported in 1990. In this study, protein production was detected after injecting reporter mRNA into mice (Pardi et al. 2018). Subsequently, a research conducted in 1992 showed that physiological responses were induced in rats after the administration of vasopressin-encoded mRNA in the hypothalamus (Figure 1) (Litvinova et al. 2023). In comparison to conventional vaccines, mRNA-based nucleic acid vaccines are anticipated to provide numerous benefits. One significant advantage lies in their differentiation from DNA vaccines, as they mitigate the risk of insertion mutagenesis and enable the modulation of expression levels of particular antigens. Second, the target of mRNA is abundant, and it is not subject to druggability of the target protein and intracellular/extracellular restrictions. Third, mRNA vaccines can be produced cell-free, bypassing the

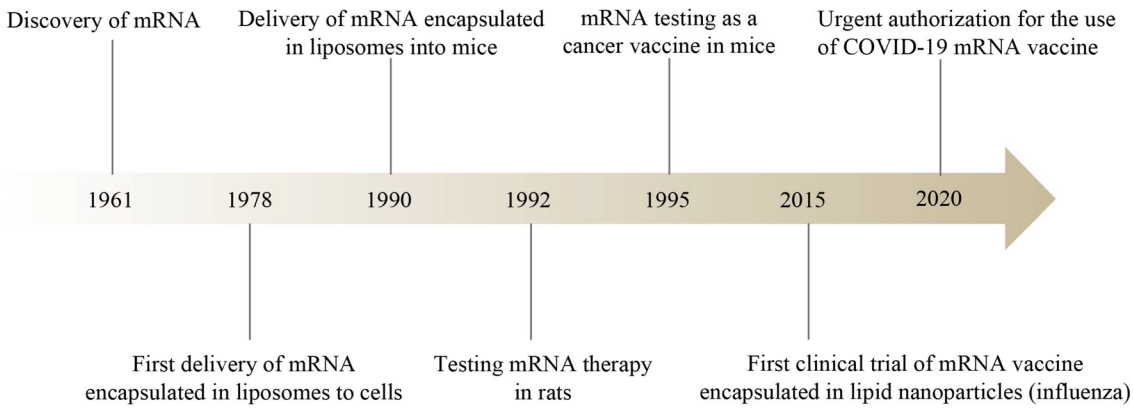

FIGURE 1 | Timeline of major events in the history of mRNA vaccine.

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challenges of chemical synthesis, and self-induced molecules are more potent. This enables fast, scalable, and cost-effective production. A single mRNA vaccine has the capacity to encode a diverse array of antigens, thereby amplifying the immune response to various pathogens. This enables the vaccine to effectively target a broad spectrum of microorganisms or virus variants using a standardized formula (Chaudhary, Weissman, and Whitehead 2021; Qin et al. 2022). The production platform exhibits strong ductility and reproducibility, making it easy to scale up production. These advantages have made mRNA vaccines a promising type of vaccine.

However, mRNA was limited by its instability, inefficient in vivo delivery, and excessive immunogenicity at its early stage, which impeded substantial investment and development in mRNA therapies. Until SARS-CoV-2 mRNA vaccine successfully broke through the technical bottleneck of vaccine preparation, enabling the rapid and cost-effective production of mRNA vaccines with excellent stability and high efficacy. It is believed that this breakthrough will benefit mRNA vaccines, with a positive impact in the future for treating tumors, autoimmune encephalomyelitis, influenza A, and many other diseases. In 2021, Nature Reviews Drug Discovery published a study analyzing 76 preventive vaccines from 31 mRNA companies worldwide. The research revealed that 22% of these vaccines are designed to prevent COVID-19 infection, while 77% are intended for other infectious diseases. Of the 32 therapeutic vaccines, 21% were associated with cancer. Among the 72 mRNA therapeutic drugs, tumor drugs represented 13%, rare disease drugs accounted for 20%, and respiratory disease drugs made up 17% (Wen, Baiping, and John 2021).

In fact, mRNA vaccines rely on somatic cells as the primary facility for naturally triggering both innate and adaptive immunity. This process facilitates post-translational modifications, leading to the complete functionality of protein products. It enables the accurate translation of polymeric and multifunctional proteins in host cells, which cannot be synthesized in folded and assembled bioreactors (Wuebben, Bartok, and Hartmann 2022). This process facilitates the transportation of proteins located in intracellular and transmembrane to their designated cellular site. mRNA vaccines have the capability to be tailored to encode specific antigens according to the distinct characteristics of the disease (Xie, Yao, and Xia 2023). From a commercial perspective, mRNA vaccines can be swiftly developed and manufactured on a large scale through cell-free processes due to their effective transcriptional response in vitro, which also offers significant cost advantages (Saraf et al. 2024). It is noteworthy that stability and immunogenicity of mRNA vaccines are limited in compared with other vaccine types, which restrict their in vivo application. However, substantial progress in mRNA modification techniques, along with advancements in the development and implementation of delivery systems, has effectively mitigated these challenges. This guarantees effective delivery and internal equilibrium (Yuan et al. 2023; Zhou and Li 2023). Furthermore, these advancements in research guarantee the preservation of mRNA integrity and the equilibrium between triggering a strong immune response and averting irreversible adverse reactions stemming from prolonged activity. The combination of mRNA with various adjuvant molecules undoubtedly establishes a robust basis for the clinical utilization of mRNA. The aforementioned reasons have positioned mRNA vaccines as a promising strategy to prevent and treat various diseases (Xie, Yao, and Xia 2023; Zhang, Tang, et al. 2023).

3 | Adjuvants in mRNA Vaccines

mRNA vaccines show significant potential in eliciting protective immune responses against pathogens. Both the antigenic component and the specific adjuvants present in mRNA vaccines have the capacity to more effectively enhance immune system stimulation (Pulendran, Arunachalam, and O'Hagan 2021; Xie, Yao, and Xia 2023). While exogenously synthesized mRNA, serving as an adjuvant, has demonstrated the potential to trigger immune responses, its effectiveness is constrained by challenges in mRNA delivery within the body and the requirement for immunostimulation (Zhang et al. 2022). Hence, vaccine adjuvants are conventionally defined as ingredients in vaccines serving as immune enhancers, which enhance the strength or induce qualitative changes in the immune response by interacting with pattern recognition receptors (PRRs) present on innate immune cells. This process ultimately augments the effectiveness of the vaccine (Reed, Orr, and Fox 2013; Xie, Yao, and Xia 2023). Furthermore, adjuvants decrease the quantity of antigen necessary for each vaccine dose and the frequency of required vaccinations. In certain cases, the stability, half-life, and immunogenicity of the antigen may be enhanced by adjuvants.

Several classic types of human vaccine adjuvants include aluminum salt adjuvants, emulsified adjuvants, TLR agonists, and granular adjuvants (Figure 2). Among these, aluminum salt

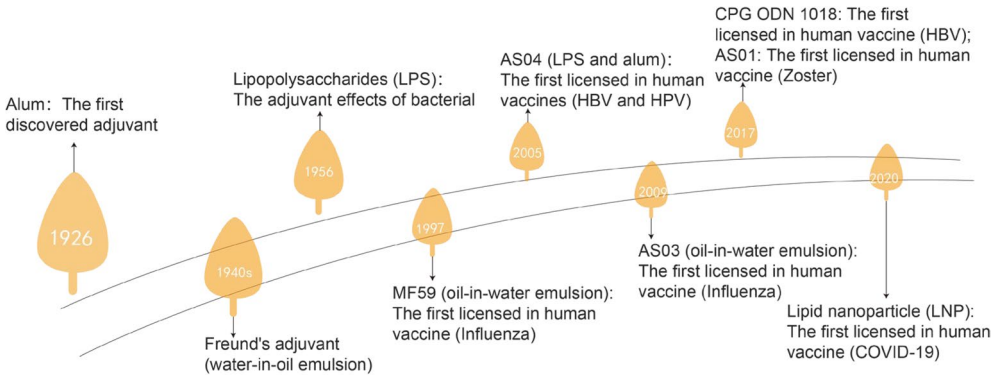

FIGURE 2 | Timeline of major events in the history of vaccine adjuvant.

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stands out as the most frequently utilized adjuvant. While classical adjuvants have been extensively utilized in various vaccines, they still present certain challenges. These include a weak effect on cellular immune response, insufficient long-term protection of immunity, and poor efficacy in the elderly. Therefore, new adjuvants need to be developed to address these problems (Orr et al. 2019). In particular, with the rapid development of nanotechnology and molecular biology, the integration of these disciplines has become essential in the design and production of antigens and adjuvant components. This integration has led to increased vaccine diversity and efficacy. Adjuvants come in various forms, such as adjuvant molecules serving as delivery vectors for mRNA, cleverly loaded or coupled to the delivery vector. The delivery vector by mimicking the size or spatial configuration of natural pathogens is engineered to specifically target innate immune cells and stimulate PRR signaling pathway to enhance a specific adaptive immune response. This mimicry can improve the uptake and presentation of antigens by APCs, thus acting as adjuvants (Bachmann and Jennings 2010; Zhao et al. 2023). In summary, current research on mRNA vaccine adjuvants mainly includes the following three forms: (1) mRNA itself has self-adjuvant characteristics; (2) Additional immunostimulants are added as adjuvant components of mRNA vaccine systems; and (3) Delivery vectors with self-adjuvant properties are used as adjuvant components of mRNA vaccines.

3.1 | mRNA With Self-Adjuvant Properties

The production process of mRNA vaccines is divided into five stages: synthesis, purification, modification (adding a "cap" and polyadenylate "tail" structure), loading of the untranslated regions at the head and tail (5′-UTR, 3′-UTR), and carrier encapsulation (Vrinda et al. 2023). In the design of vaccines, adjuvants are considered to enhance the immune response. In 2004, scientists confirmed that foreign RNA molecules, through the synthesis of mRNA in vitro served as vaccine adjuvants, can induce immune responses in mammalian cells (Xie, Yao, and Xia 2023). Studies have shown that nucleotides in mRNA can exert adjuvant activity through their inherent immunostimulatory properties (Kobiyama and Ishii 2022). For example, exogenetic mRNA, especially unmodified mRNA, can activate the innate immune system by binding to PRRs (Katalin et al. 2005). Single-stranded RNA (ssRNA) activates the myeloid differentiation marker 88 (MyD88) pathways via recognizing TLR7 and TLR8 in endosomes (Järver et al. 2018). Double-stranded RNA (dsRNA) is reported that can be recognized by TLR3, RIG-I, melanoma differentiation-associated gene 5 (MDA5), and other molecules, which stimulate the production of pro-inflammatory cytokines (such as TNF-α, IL-6, and IFNα/β), thereby amplifying the protective immune response (Wang et al. 2019). TLR is a single-receptor transmembrane superfamily consisting of 11 family members such as TLR1, TLR2, TLR4, TLR5, TLR6, and TLR11 located on the cell surface, while TLR3, TLR7, TLR8, and TLR9 are located in the endosome/lysosome compartment (Figure 3). It is expressed in many tissues and cells. TLR primarily recognizes various types of pathogen-associated molecular patterns (PAMPs) (such as lipopolysaccharides [LPS], sodium urate crystals, flagellin, peptidoglycan [PGN], viral double-stranded RNA, and unmethylated CpG motifs in bacterial DNA) through germline coding patterns, and induces inflammatory and immune responses in the body within minutes to hours. TLR1/2 and TLR2/6 recruit TIRAP (Mal) and MyD88 and then activate NF-κB; TLR3 induces to release the type I IFN and inflammatory cytokines by activating the TRIF-dependent pathway; TLR4 recruit the TIRAP and MyD88 to activate the early-phase of NF-κB by complexed with MD2. Then, it was internalized and retained in the endosome, activating IRF3 and late-phase NF-κB to induce type I IFN by recruiting TRAM and TRIF; TLR5 recruits MyD88 to activate NF-κB; TLR7/9 located in the endosome recruits MyD88 to release type I IFN and inflammatory cytokines by activating TRAF3 and NF-κB (Aga Syed and Saniya 2021; Barton and Kagan 2009; Kawai and Akira 2010; Lee, Avalos, and Ploegh 2012; Payam, Herney Andrés, and Tomasz 2021).

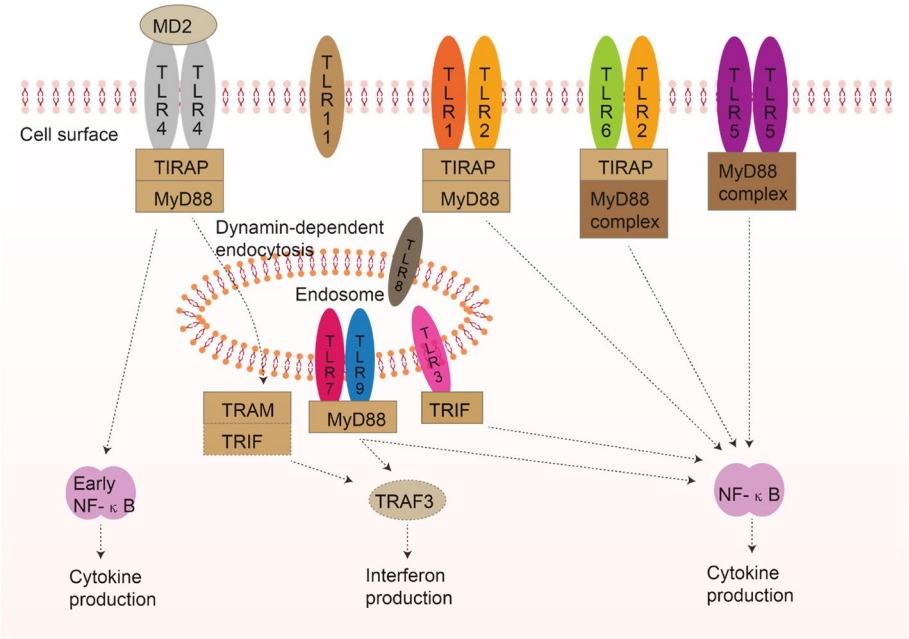

FIGURE 3 | PAMP recognition by TLRs located in cell surface and intracellular.

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In general, mRNA vaccines work through humoral immunity, cellular immunity, and self-adjuvant effects as described above. However, unmodified mRNA can lead to excessive inflammation. In addition, by-products produced during mRNA production can also be recognized by TLRs in the cytoplasm, activating interferon signaling pathways and producing various pro-inflammatory factors. This phenomenon illustrates the self-adjuvant effects of mRNA vaccines (Whitley et al. 2022). It is suggested that self-adjuvant mRNA vaccines can be established by integrating adjuvants into the single-stranded mRNA, without compromising its capacity to generate antigen proteins. Among them, Tockary et al. (2023) developed short dsRNA targeting the natural immune receptor RIG-I. They loaded it onto the mRNA single strand via hybridization, resulting in an mRNA with a comb-shaped structure that can efficiently stimulate RIG-I. The comb mRNA generated through this method effectively activates DCs, which are crucial for enhancing vaccine effectiveness.

3.2 | Exogenous Immune Stimulants

mRNA-based therapeutics and vaccines have great development potential, and synthetic mRNA vaccine technology is gradually being applied in clinical settings. However, the intrinsic immunostimulating activity of mRNA synthesized in vitro is low due to its toxicity and its capacity to enhance antigen production. mRNA requires base modification to minimize innate immune recognition, resulting in only weak self-adjuvant activity (Jilian et al. 2022). Consequently, the inclusion of adjuvants is essential to augment the immune response. In this scenario, adjuvants are typically introduced into mRNA systems in two forms: as exogenous materials or "built-in" adjuvants (Pulendran, Arunachalam, and O'Hagan 2021). Such "built-in" adjuvants by incorporated into vaccine system include nucleic acids presenting in both viral and non-viral vector vaccines, lipids recognized as PAMPs (Kobiyama and Ishii 2022). These constituents stimulate PRRs present in APCs or TLRs and cGAS in a manner dependent on their structure, initiating specific immune responses from T and B cells (Kong et al. 2023; Jilian et al. 2022). Nevertheless, the mRNA could potentially undergo "immunosilencing," resulting in a diminished self-adjuvant effect. The issue at hand can potentially be addressed by mRNA vaccine delivery, which offers a promising solution through the reduction of DC activation and the consequent activation of T cells.

LNPs are one of the most utilized delivery system for mRNA vaccines due to their biocompatibility, stability, and capacity to safeguard mRNA, which usually can be divided into cationic LNPs and ionizable LNPs according to the lipid components, surface charges, and functional groups (Hou et al. 2021; Oberli et al. 2017). The cationic LNP encapsulates the negatively charged mRNA through electrostatic interactions facilitated by its positively charged lipid components. This mechanism also facilitates the transportation of mRNA to target cells (Simon et al. 2016; Zhang et al. 2021). Another type of LNP is ionizable LNP, which is composed of a double layer of ionizable lipids surrounding an mRNA core. The double layer of ionizable lipids plays a crucial role in facilitating the endosomal release of mRNA cargo and facilitating cytoplasmic transport, which are essential for achieving efficient protein expression. Hence, enhancing the immunogenicity of modified mRNA–LNP vaccines poses a challenge in certain application scenarios (Figure 4a). For instance, a lipid–PEG was employed as a carrier system for ovalbumin-encoded mRNA and TLR7/8 agonist R848 modified by palmitic acid (C16-R848, serving as the adjuvant). The encapsulated C16-R848 activity was preserved, and the mRNA transfection efficiency was notably enhanced by the presence of the lipid–PEG shell (Islam et al. 2021). Compared with the unadjuvanted mRNA vaccine NPs, the C16-R848 adjuvant-pulsed mRNA vaccine can stimulate both innate and adaptive immune responses simultaneously, resulting in fewer toxic side effects. This method is a good combination of adjuvant and delivery vector to create a self-adjuvant nanocarrier, enabling the prevention and treatment of H5N1 influenza virus.

Similarly, Brito et al. (2014) employed a cationic nanoemulsion (CNE) to construct the mRNA nanovaccine for amplification (Figure 4b). The self-amplified mRNA delivered by CNE was taken up by immune cells, and was well tolerated and induced immunogenicity in various animal models. Of course, in addition to cationic nanoemulsions, polymers such as polyethylenimine (PEI) possess strong positive properties and can effectively bind with negatively charged mRNA through electrostatic interactions. PEI also exhibits good permeability through biological cell membranes, making it widely utilized in the development of nanodelivery systems. For example, Démoulins et al. (2021) concurrently administered RepRNA and a double (3′, 5′) cyclic dimer adenosine phosphate (c-di-AMP) adjuvant packaged in viral replicant particles (VRP) with PEI to formulate PEI-based polymer nanophosphate vaccines (Figure 4c). The research revealed that pigs administered with VRP in conjunction with PEI-delivered RepRNA exhibited a robust humoral and cellular response. Hence, the choice of suitable synthetic delivery vectors continues to offer the potential for expeditious vaccine development.

3.3 | Components of the Delivery System

With the chemical modification of N1-methylpseudouridine (m1ψ) proposed by Drew Weissman and Katalin Kariko, along with the invention of the delivery vector LNP, the challenges in mRNA vaccine advancement have been successfully addressed, specifically those associated with cationic lipids. Certain lipid constituents within the LNP mRNA system vaccines, such as the ionized cationic lipid DLinDMA, can function as adjuvants exerting to induce a potent TFH cell response and germinal center (GC) B cell response, resulting in abundant neutralizing antibodies (Hongxia et al. 2021; Li et al. 2023; Vimal Kumar et al. 2017). For example, calcium phosphate, a component in the LNP, severs as a synthetic nanoscale Ca^{2+} reservoir, which aids in the maturation of DCs by enhancing the levels of Ca^{2+} in the cytoplasm and increasing the expression of costimulatory molecules like CD80 and CD86 (Cao et al. 2024). Furthermore, the cationic lipid-like delivery vector can induce the release of inflammatory cytokines, and boost the production of costimulatory molecules via the TLR4 pathway upon the delivery of mRNA into the intracellular context (Xie, Yao, and Xia 2023).

On the other hand, other nanoparticles themselves show the potentiality serving as adjuvants in vaccines, such as

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molecular PRR agonists, self-adjuvant polymer/lipid materials, inorganic materials, and bio-derived materials, which can effectively stimulate innate immune responses (Figure 5, Table 1) (Li et al. 2021; Liao et al. 2022; Luo et al. 2017; Miao et al. 2019; Xie et al. 2022). Another noteworthy feature is the relatively simple chemical composition of these self-adjuvant materials. This simplicity contributes to the consistency in adjuvant manufacturing, facilitates quality control, and ultimately lowers the expenses linked to mRNA vaccine production. Moreover, their physical and chemical properties offer advantages that can streamline the vaccine manufacturing process and potentially mitigate side effects associated with vaccine formulation management. Superior therapeutic properties have been demonstrated in comparison to conventional adjuvants. Self-adjuvant mRNA nanovaccines are composed of nanomaterials with intrinsic immunostimulating activity. This composition ensures high spatial and temporal synergies, optimal pharmacokinetic properties, and immune enhancement effects, which collectively improve the therapeutic effect (Liu et al. 2021; Luo et al. 2020). For instance, it has been documented that protamine has the capability to create a complex with mRNA, which can activate B- and T-cell-dependent responses via the TLR7/8 pathway against influenza A or tumors (Fotin-Mleczek et al. 2011). In contrast to protamine, the cholesterol-modified cationic peptide DP7 (VQWRIRVAVIRK) is also reported to boost the

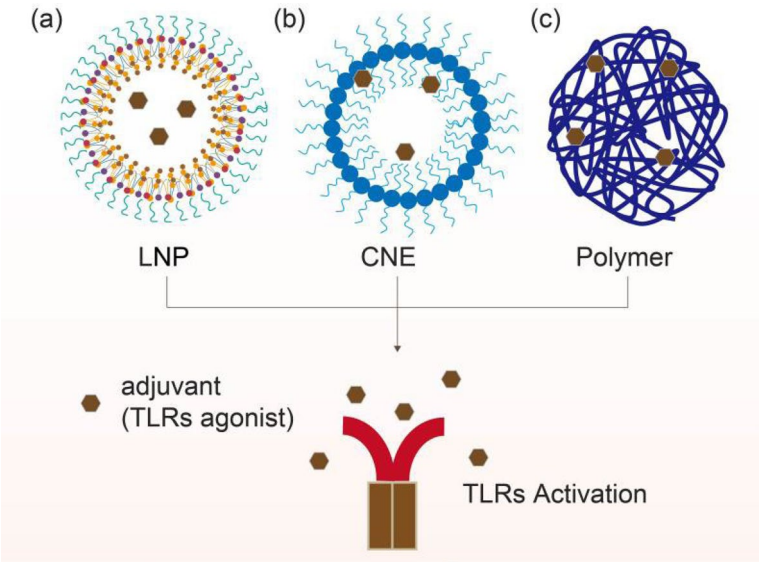

FIGURE 4 | Exogenous adjuvant molecules combine with delivery system to exert self-adjuvant effect in mRNA nanovaccine.

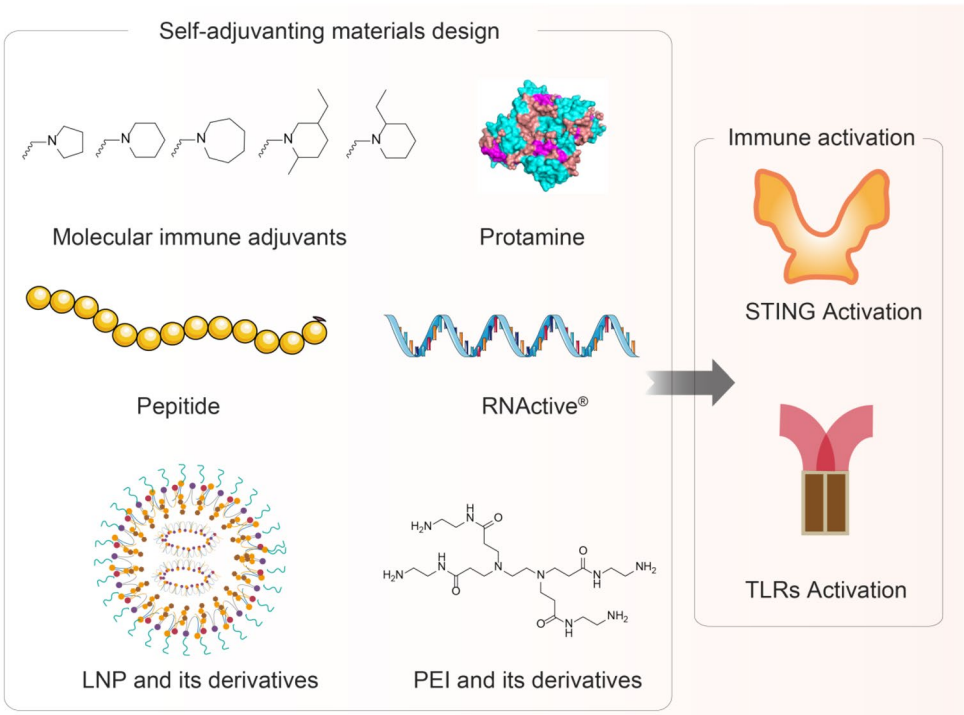

FIGURE 5 | Self-adjuvant mRNA delivery vector and immune activation mechanism.

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TABLE 1 | mRNA self-adjuvant vaccine and its application.

Self-adjuvant component	Immune pathway	Applications	References
Peptide from CFP-10 protein	TLR-2	Tuberculosis	Mishra et al. (2023)
C3d and ionizable lipid	Potent T-cell responses	SARS-CoV-2	Li et al. (2023)
Lipidoid-substituted LNPs	TLR-7/8	SARS-CoV-2	Han et al. (2023)
Self-replicative RNA	Specific T-cell responses	SARS-CoV-2	Akina et al. (2022)
Lipid-like material	TLR4	Tumor	Zhang et al. (2021)
A tertiary amine (PC7A)	STING	Tumor	Li et al. (2021)
Heterocyclic lipids	STING	Tumor	Miao et al. (2019)
Polyethylenimine (PEI) diverse	Potent humoral and cellular immune responses	Pandemic	Li et al. (2017)
mRNA encoding influenza A hemagglutinin	TLR-7/8	Influenza A	Edwards et al. (2017)
RNActive	Potent and boostable adaptive immunity	Tumor	Kowalczyk et al. (2016)
RNActive	T and B cell responses	Prostate cancer	Kubler et al. (2015)
RNActive	Enhance immune responses	Lung cancer	Martin et al. (2014)
Cationic lipid	Evoking functional T cell responses	Cancer and HIV	Pollard et al. (2013)
Lipid-derived nanoparticles	TLR-7/8	Influenza virus	Nguyen et al. (2012)
Protamine	TLR7	Tumor	Fotin-Mleczek et al. (2011)

immune response by activating the TLR2-MyD88-IKK-NF-κB pathway. This characteristic renders it a potent adjuvant for tailored cancer vaccines. In brief, immunostimulants that exhibit adjuvant properties in mRNA vaccines consist of proteins, functional peptides, ligands that bind to native immune receptors, and agonists of native immune cells. These supplementary immune stimulants generally operate by triggering TLR signaling (Xie, Yao, and Xia 2023).

4 | Application of Self-Adjuvant mRNA Vaccine in Disease

4.1 | Cancer

Vaccines aimed at preventing infectious diseases were one of the most medical developments in the 20th century. So far, the vaccination has evolved significantly, extending beyond prevention to include the emergence of therapeutic vaccines. Encouraged by this, the development of cancer vaccines aimed at activating antitumor immunity is expected to provide a new strategy against tumor (Lin et al. 2022). In particular, mRNA vaccines have shown outstanding performance against SARS-CoV-2, which has been widely recognized and sought after by the scientific community and industry. At the same time, mRNA vaccines are an essential form of cancer vaccine, and the application of mRNA vaccine technology to develop new therapeutic and preventive tumor vaccines also brings new opportunities for tumor treatment. Effective tumor mRNA vaccines translation (such as functional proteins, tumor antigen) can generate active specific immunity to potentially inhibit or eliminate tumor growth, recurrence, or metastasis and intervene in the initiation phase of the tumor immune cycle (Figure 6) (Jagmohan, Wilbur, and Adam 2021; Jian et al. 2022; Salleh, Norazmi, and Deris 2022; Zhang and Xia 2021). They have the capacity to encode and express a diverse range of tumor-associated antigens (TAAs), tumor-specific antigens (TSAs), and the respective cytokines within tumor cells or microenvironments. This process triggers both humoral and cellular immunity, thereby improving their suitability for patients with different tumor types (Fan et al. 2023; Liu et al. 2023). This leads to a reduction in the number of repeated treatments in comparison with conventional cancer therapies, thereby enhancing the efficacy of the treatment. Furthermore, mRNA that encodes chimeric antigen receptors (CAR) or T cell receptors (TCR) can effectively target proteins due to their high transfection rates. CAR T products or TCR-engineered T cells produced ex vivo or in vivo are anticipated to be employed in the treatment of cancer (Chaudhary, Weissman, and Whitehead 2021). However, currently approved adjuvants for therapeutic mRNA vaccines are unable to stimulate robust antitumor cell immunity in the body.

Compared with conventional infectious disease vaccines, two fundamental obstacles must be overcome to achieve high efficacy of tumor vaccines: the immunogenicity and tumor immunosuppression. To overcome these two obstacles, it can begin with vaccine combinations and combined tumor immunization. Vaccine formulation involves the dosage form, which includes carriers, solvents, immune activators, agents that inhibit inhibitory signals, and agents that enhance the microenvironment.

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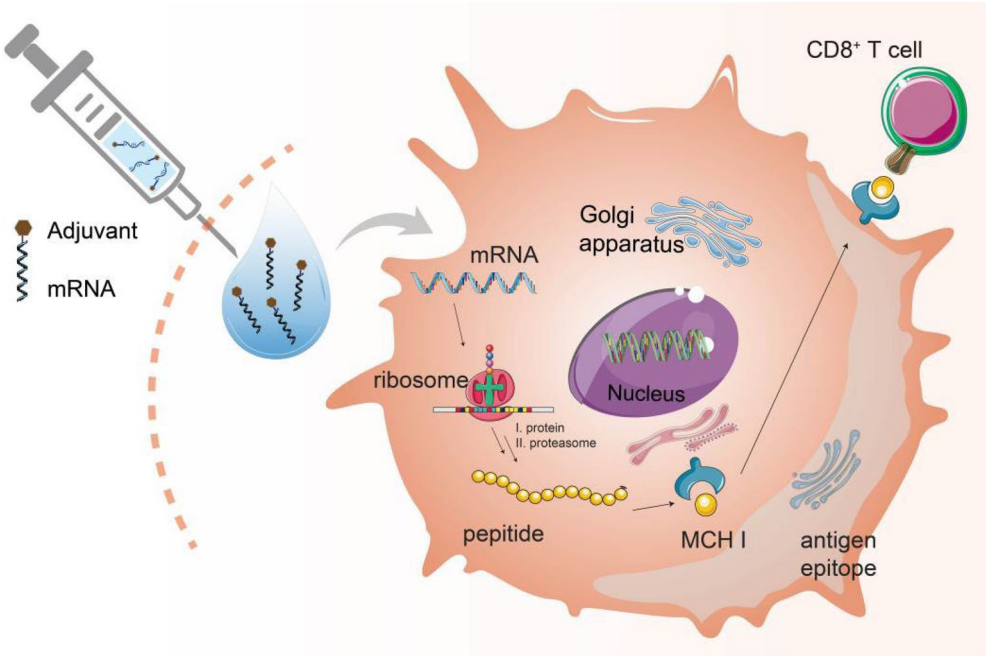

FIGURE 6 | mRNA vaccine activation mechanism (Lorentzen et al. 2022).

Combined tumor immunization is a treatment for external application. Adjuvants are utilized in the formulation and administration of personalized tumor vaccines. Carriers that either promote or inhibit inhibitory signals and alter the microenvironment are all considered adjuvants, and they can be used in combination or individually. Adjuvants, as the principal constituents of tumor vaccines, primarily serve to augment the magnitude, velocity, and persistence of the immune reaction elicited by the vaccine. Cancer vaccines operate through the recognition and elimination of cancer cells facilitated by vaccine adjuvants. This process involves the uptake and display of tumor antigens by APCs, leading to the activation of T cells specific to the antigens (Xuanbo Zhang, Ni, and Chen 2023).

Among these approaches, direct vaccination using mRNA molecules encoding TAAs is undoubtedly a good choice in cancer immunotherapy because of two advantages: (1) mRNA can induce adaptive immunity as an antigen source itself; (2) mRNA can stimulate innate immunity when it binds to PRRs. Although it achieving these two features above, its translatability still will be inhibited by the mRNA complex required for immunostimulatory activity. Therefore, the development of a sophisticated carrier with immunostimulatory activity, particularly possessing adjuvant properties, holds significant importance. Based on this, the Fotin-Mleczek et al. (2011) introduced an mRNA-based two-component tumor vaccine designed to facilitate antigen expression and immune stimulation through TLR7 mediation. TLR agonists trigger immune stimulation and enhance the immunostimulatory activity of mRNA molecules. Such complexes have the potential to improve the efficacy of vaccines and obviate the necessity for exogenous adjuvants. In this strategy, protamine serves as both an mRNA delivery carrier and an adjuvant molecule aimed at augmenting the immune response to the vaccine.

In addition to protamine, LNPs are considered as one of the most advanced forms of vaccine carriers, which hold significant clinical transformative potential due to their adjuvant properties. In this regard, Zhang et al. (Zhang and Xia 2021; Zhang et al. 2021) developed an mRNA nanovaccine utilizing C1 LNPs following the evaluation of various cationic LNPs materials. This approach successfully facilitated the delivery of mRNA to APCs and elicited significant T cell responses (Figure 7a). Moreover, C1 LNPs themselves induce the secretion of IL-4 in DC cells, by activating the TLR12 signaling pathway. C1 mRNA nanovaccines with self-adjuvant properties exhibit notably efficient antigen expression. The findings illustrate substantial antitumor effectiveness in both tumor prevention and therapeutic vaccine scenarios, potentially laying the groundwork for the progression of cancer immunotherapies directed at diverse tumor varieties. Furthermore, there are reports indicating that immune-stimulating cytokines are being explored for their potential application as adjuvants in cell-based therapies and vaccines. The utilization of this option in preclinical studies is highly attractive due to its ability to attract DC to the injection site, enhance DC maturation, and facilitate antigen presentation. These mechanisms collectively augment immune responses specific to cancer (Gupta and Wahi 2022).

TLR agonists initiate immune stimulation and improve vaccination efficacy. The most commonly used TLR agonists include polyinosinate-polycytidylic acid, carboxymethyl cellulose, monophosphoryl lipid A, flagellin, imiquimote, guanidine motif modified branched PEI, and CPG-ODN. Another class of potential cancer vaccine adjuvants includes STING, which embedded in the endoplasmic reticulum that activates type I IFN in response to DNA within the cell as a transmembrane protein (Luo et al. 2017). cGAS, comprising a nucleotide transferase domain and two major DNA binding domains, binds to cytoplasmic DNA along with the conformational change producing the cGAMP molecule, which are the natural ligands for STING via hydrophobic interactions and hydrogen bonding leading to the activation of STING (Yu and Liu 2021). The carboxyl terminus

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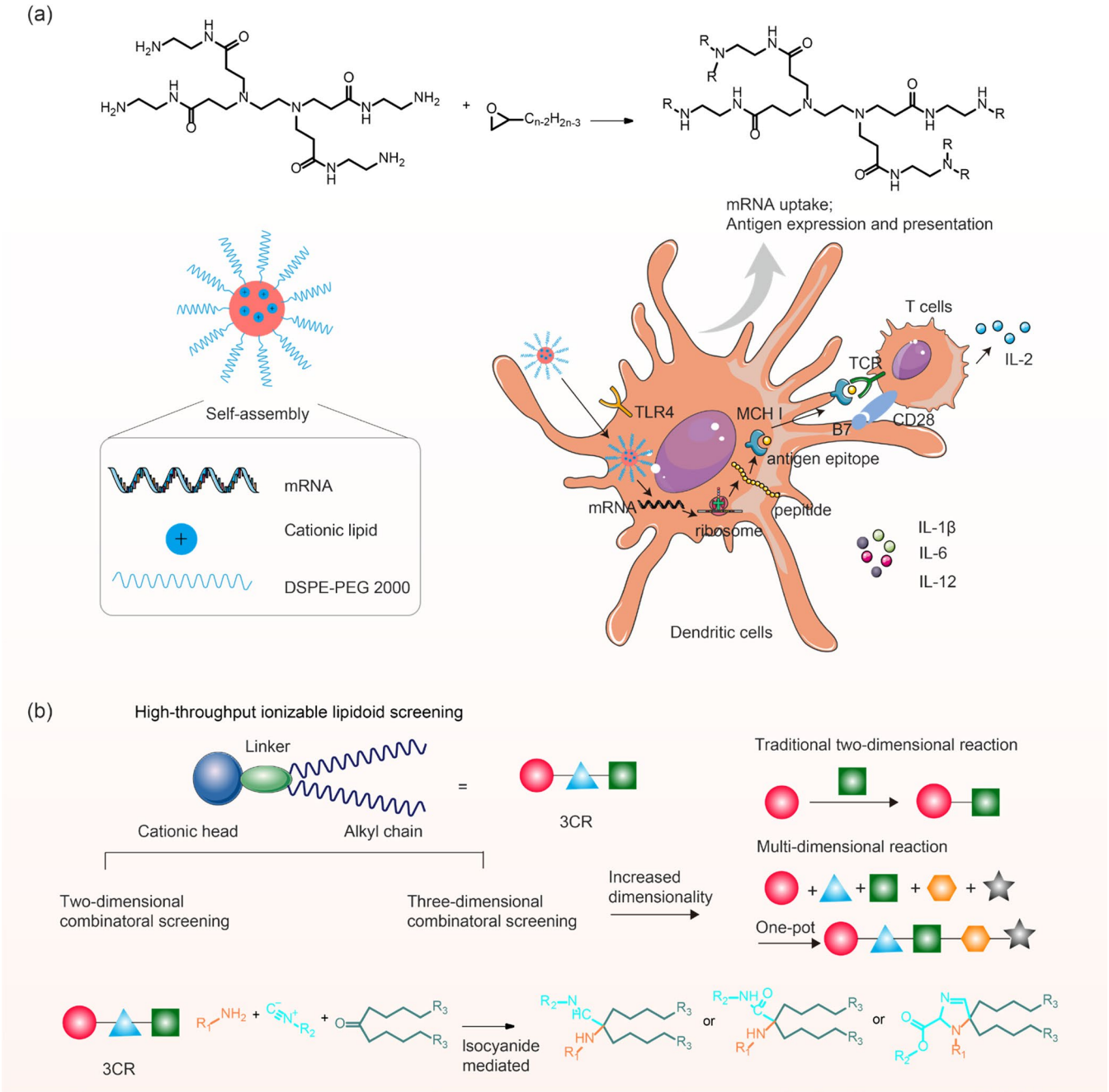

FIGURE 7 | Design, preparation, and mechanism of self-adjuvant mRNA nanovaccines in cancer of. (a) T cell activation-based screening of mRNA-delivering lipid-like materials. Reproduced with permission from Zhang et al. (2021), Copyright 2021. (b) Isocyanide-mediated 3-CR for high-throughput synthesis of lipidoids (Miao et al. 2019).

of STING recruits and activates TBK1, as well phosphorylates the transcription factor IRF3, which then dimerizes and enters the nucleus, being with the production of type I IFN, MHC I and co-stimulatory molecules such as CD86 (Figure 8) (Civril et al. 2013; Jing et al. 2023; Xinyuan et al. 2023).

However, the exogenous delivery of STING agonists, which exert their functions by penetrating the cell membrane, is hindered by the poor pharmacokinetics and physicochemical properties (Zhivaki and Gosselin 2023). Activation of the STING pathway may trigger apoptosis in T cells unlike macrophages and DC due to their higher levels of STING expression (Luo et al. 2017, 2020; Su et al. 2022). In comparison to TLR agonists, STING activators, such as small molecules or polymers, induce comparatively lower levels of both local and systemic inflammation. For example, Miao et al. (2019) created a combinatorial library of ionizable lipid-like materials through the utilization of a three-dimensional multi-component reaction (3-CR) system. This system serves the dual function of facilitating mRNA delivery and inducing robust and targeted immune activation via the STING pathway (Figure 7b). In this context, the lipid head group assumes a essential role in the carrier system, facilitating the modulation of the immunostimulatory effect through the utilization of these lipids. Furthermore, it has been demonstrated that lipids with cyclic amino head groups can activate the STING pathway autonomously from MYD88, particularly in relation to TLR7 or

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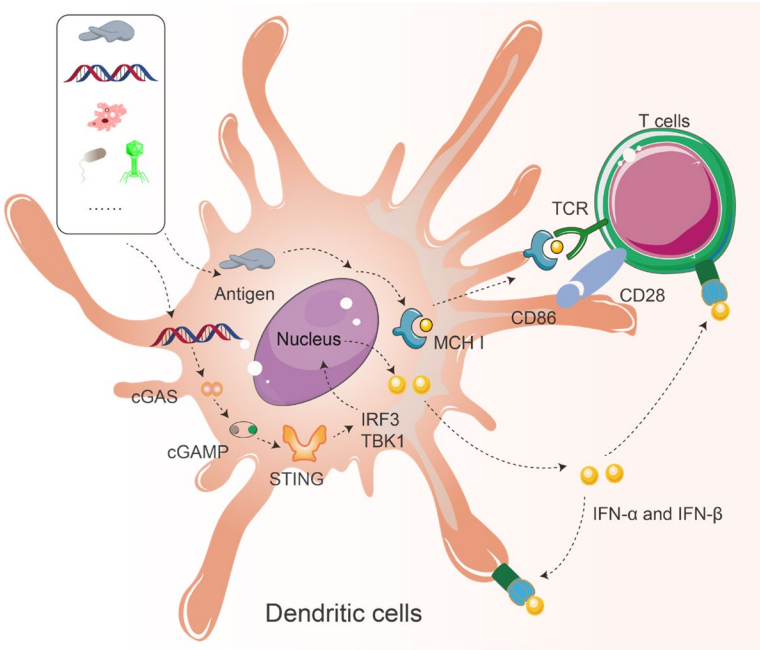

FIGURE 8 | The cGAS pathway activated by DNA and other stimuli.

8, as well as RLR (RIG-I and MDA5). This lipid-adjuster-assisted mRNA vaccination with LNPs induces APCs to mature intracellularly through the STING pathway rather than via TLRs. This leads to restricted systemic cytokine expression, leading to enhanced antitumor efficacy.

4.2 | Infectious Disease

The nucleoside-modified mRNA effectively enhances the stability of mRNA vaccines, reduces immunogenicity, and avoids uncontrollable inflammatory reactions. However, this modification also greatly weakens the inherent immune response of mRNA. For example, Han et al. (2023) have formulated mRNA–LNP vaccines by incorporating an adjuvant lipid as part of the LNP component to augment adjuvant properties (Figure 9a). Their study demonstrated that ionizable lipids containing adjuvant lipidoid are advantageous in improving mRNA delivery, leading to a predominant TH1 cellular immune response, as well as a vigorous B cell response and sustained cell responses against various SARS-CoV-2 pseudovirus variants. Significantly, this self-adjuvant lipid substitution strategy has proven effective in mRNA-LNP vaccines that are clinically relevant, showcasing its potential for translation into practical applications.

Similarly, Li et al. (2023) modified both the mRNA itself and its delivery vector LNP to improve the efficacy of mRNA vaccines in inducing immunity. A "self-adjuvant" mRNA vaccine has been developed through a dual strategy (Figure 9b). The vaccine comprises LNPs that are enveloped with mRNA-encoded antigens designed to facilitate effective mRNA delivery and to boost the innate and adaptive immune responses. They screened a library consisting of 480 biodegradable ionizable lipids to enhance the effectiveness of vaccines. The vaccines described utilize head-based adjuvants that are enhanced with cycloamines and adjuvants through the fusion of an mRNA-encoded antigen with C3 complement protein as a natural adjuvant Experiments confirmed that the new mRNA vaccine increased the level of neutralizing antibodies produced by the mice by more than 10 times and also elicited a stronger T-cell response. The researchers further inoculated the novel RNA vaccine platform intranasally and observed a robust immune response similar to that elicited by injectable vaccination. This suggests that the vaccine can provide the same level of immune protection through respiratory inhalation, giving it a natural advantage against diseases transmitted through the respiratory tract. Multiadjuvant mRNA vaccines can enhance the effectiveness, safety, and convenience of mRNA-based immunization.

Mycobacterium tuberculosis (Mtb) is the pathogen responsible for causing tuberculosis. Despite its partial efficacy against tuberculosis, the Bacille Calmette-Guérin (BCG) vaccine remains the most commonly used vaccine. Mishra et al. (2023) developed a synthetic mRNA vaccine, SelmRp25-C5 that expresses the p25 epitope from the Mtb-derived Antigen-85B and the TLR2-stimulating C5 adjuvant peptide from the CFP-10 protein. The protective effect was associated with an increase in Ag85B-p25 specific cells and elevated levels of IFN-γ and IL-2 secretion effectors. mRNA vaccine expressing the C5 adjuvant peptide acted as a self-adjuvant, enhancing innate adaptive immunity to provide protection. Extensive research has been undertaken to enhance the design strategy of mRNA vaccines directed towards the H5N1 influenza virus. Studies have concentrated on investigating the potential enhancements in immune efficacy of mRNA vaccines through various antigen designs and adjuvants. To this end, Edwards et al. (2017) employed a range of combinatorial analysis methods to introduce an innovative translational strategy for the preclinical assessment of the fundamental mechanisms of mRNA vaccine self-adjuvant in vitro human models known as MIMIC, which was employed to assess the innate responses elicited by various doses of mRNA vaccines encoding

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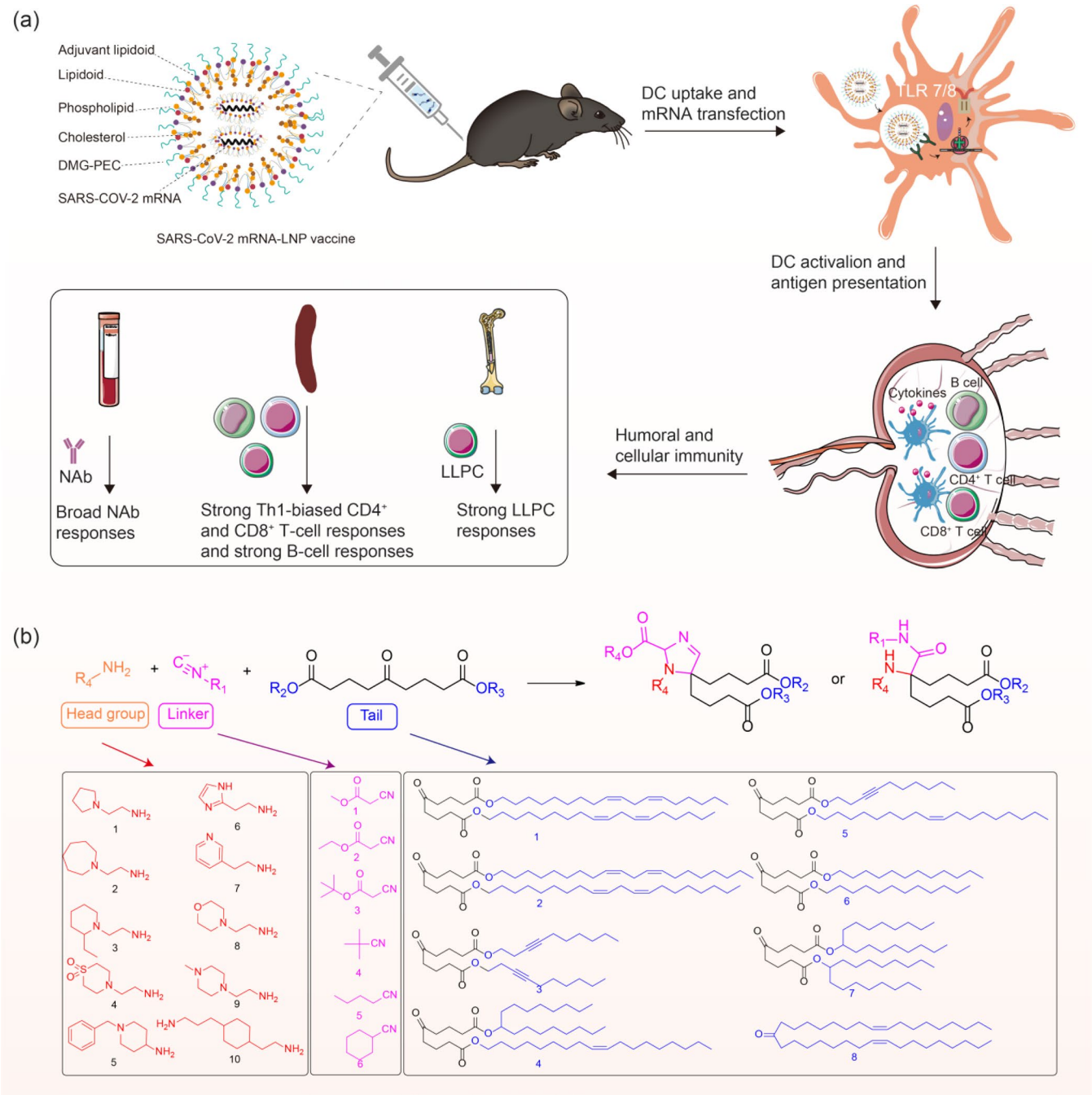

FIGURE 9 | Design, preparation, and mechanism of self-adjuvant mRNA nanovaccines in SARS-CoV-2. (a) Adjuvant lipidoid-substituted SARS-CoV-2 mRNA-LNP vaccine and its proposed mechanism of action to elicit potent adaptive immunity (Han et al. 2023). (b) High-throughput synthesis and screening of biodegradable cyclic ionizable lipids synthesized by Ugi-3CR (Li et al. 2023).

influenza A hemagglutinin in genetically uniform mice. The results revealed that the immune response elicited by the mRNA vaccine in humans closely resembles that observed in mice. The mechanism operates through the utilization of cellular RNA sensors to promote the maturation and activation of immune cells, as well as the production of cytokines and chemokines. The transformation approach employed in this preclinical assessment has significantly altered the basic mechanism of mRNA vaccine self-adjuvantation, thereby elucidating the operational mechanism of the vaccine in both human and murine models.

4.3 | Other Diseases

The Zika virus is a mosquito-borne virus initially discovered in rhesus monkeys in Uganda. Notably, the majority of individuals infected with the Zika virus are asymptomatic. Currently, there is no vaccine available for the prevention or treatment of Zika virus infection. Hence, there is ongoing active pursuit in the development of Zika vaccines. To this end, various polymer-based nanovaccine systems employed for intranasal administration to achieve rapid immune protection. These systems have shown the ability to confer self-adjuvant activity. Based on this

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discovery, Li et al. (2017) employed PEI, known for its potent mucosal adjuvant properties, to develop an intranasal self-adjuvant mRNA nanovaccine. This approach markedly improves the efficacy of intranasal mRNA vaccines via enhancing the ability to the lymph nodes, consequently boosting the immune response. The optimization of the PEI chemical structure to improve its adjuvant properties while reducing its toxicity poses a significant challenge in the field. To achieve this goal, cyclodextrin (CD) was employed in conjunction with PEI600 or PEI2k for the synthesis of CP (CD-PEI) polymers. The chemical composition of PEI was found to be a critical factor influencing the migration of nanocomplexes formed by PEI and mRNA to the lymph nodes and inducing immune responses. For instance, CP600, CP2k, and PEI25k were screened as delivery systems for intranasal mRNA vaccines. This screening involved evaluating their lymph node migration and immune response capacities. Among them, the mRNA vaccine based on CP2k exhibited significantly higher in vitro transfection efficiency, improved migration to lymph nodes, and the ability to stimulate DC maturation in vivo. This resulted in an effective humoral and cellular immune response, showing reduced local and systemic toxicity compared with PEI25k/mRNA. The findings presented in this study emphasize the potential of CD-PEI2k/mRNA nanocomplexes as self-adjuvant vaccine delivery vectors for efficient transport to lymph nodes.

5 | Conclusion

5.1 | Adjuvants in Conventional Nanovaccines

mRNA vaccines have become a focal point of investigation in the field of disease prevention and therapy, showing considerable potential in combating infectious diseases and managing cancer. They have emerged as a robust alternative compared with conventional vaccines, showcasing significant advantages of efficiency, safety, effectiveness, and their capacity to expedite clinical development and enable cost-effective manufacturing. However, with the exception of COVID-19 mRNA vaccines, there have been limited approvals for mRNA vaccines for disease treatment thus far. Conventional nanovaccines require simultaneous loading of antigens and adjuvants, which complicates the design process. Classical adjuvants have demonstrated satisfactory biosafety to date. The utilization of established research and development methodologies, in conjunction with rigorous production conditions and suitable equipment, offers a relatively uncomplicated framework for the creation of innovative vaccines. However, the immunostimulating potential of conventional adjuvants remains constrained. Therefore, there is an urgent necessity to discover novel adjuvants.

5.2 | Novel Adjuvants in Nanovaccine

The extensive overlap of nanotechnology with materials science, immunology, and biomedicine has accelerated the rapid development of nano-vaccines, nano-adjuvants, and nano-delivery systems. Despite the promising strategies proposed for these adjuvant materials, challenges exist in elucidating the underlying mechanisms and effectively translating them into clinical practice. Most notably, while vaccine adjuvants enhance the effectiveness and duration of the immune response, it is essential to recognize that the primary objective in designing adjuvant materials is not exclusively centered on attaining strong and sustained immune stimulation. Inappropriate immune stimulation can result in adverse effects, such as granuloma formation, as well as induce immune tolerance and T cell depletion, thereby significantly reducing the efficacy of immunotherapy. Therefore, the formulation of adjuvants should consider the frequency of immune stimulation. To prevent immune tolerance, it is crucial to adjust the frequency of administration or release kinetics according to long-term experimental observations. In addition, the physical and chemical variances among the constituents of the "antigen + adjuvant" approach frequently result in co-delivery challenges, including adjuvant leakage and antigen inactivation. These complications may impede the scalability of production, sterilization, and storage processes.

5.3 | Self-Adjuvant Nanovaccine

To this end, the approach of incorporating adjuvant functions into the delivery system to establish an effective, multifunctional self-adjuvant vaccine platform has been developed, which exhibited some unique properties as follows: (1) Compared with "antigen + adjuvant" form, they are able to activate some specific innate immune signaling pathways (e.g., STING, TLR, and NLR). (2) These self-adjuvant materials have a relatively simple chemical structure, which conceptually improves the repeatability of adjuvant manufacturing and quality control. (3) The physicochemical advantages of these synthetic materials can simplify the process of designing and preparing vaccines. (4) The spatio-temporal coordination between the vaccine and adjuvant was effectively achieved simultaneously.

5.4 | Outlook

In recent years, there has been a growing interest in the investigation of nanomaterials with the potential to function as mRNA self-adjuvant immune carriers and mRNA nanodelivery carriers. Such an approach significantly simplifies the design and formulation of mRNA vaccines. Concurrently, it enables precise spatial control of antigen and adjuvant stimuli. To address the undesired increase in systemic inflammatory cytokine levels associated with the immunogenicity of mRNA vaccines. For example, a range of self-adjuvant mRNA nanovaccines incorporating lipid materials, PEI, protamine, chemical small molecules, polypeptides, and their derivatives have been consistently reported in the literature. These vaccines have the capacity to trigger strong antitumor immune responses, without any detectable toxicity when incorporating mRNA encoding tumor antigens. The self-adjuvant mRNA vaccine system is anticipated to enhance the safety, effectiveness, and convenience for infectious diseases and other medical conditions. It also holds the potential to improve mass production and expand its clinical application in the future.

Author Contributions

Huaibin Yu: writing – original draft (lead). **Yafang Lu:** investigation (equal), methodology (equal). **Zhuorong Miao:** writing – original draft (supporting). **Zhengbao Zha:** conceptualization (equal), supervision

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(equal), writing – review and editing (equal). **Shaoqin Liu:** conceptualization (lead), supervision (equal), writing – review and editing (equal).

Acknowledgments

We thank Henan Key Science and Technology Research (242102311235), the China Postdoctoral Science Foundation (2024M754162) for financial support.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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